

Homework 04

```
if (!requireNamespace("glmnet", quietly = TRUE)) install.packages("glmnet")
if (!requireNamespace("tidyr", quietly = TRUE)) install.packages("tidyr")
if (!requireNamespace("dplyr", quietly = TRUE)) install.packages("dplyr")
if (!requireNamespace("ggplot2", quietly = TRUE)) install.packages("ggplot2")
if (!requireNamespace("knitr", quietly = TRUE)) install.packages("knitr")

library(glmnet)
library(tidyr)
library(dplyr)
library(ggplot2)
library(knitr)
```

Question 1: Signal strength and correlated substitutes

```
set.seed(43246)

n      <- 120
p      <- 10
n_rep  <- 200
lam_grid <- c(0.08, 0.12, 0.18, 0.24)
n_lam  <- length(lam_grid)

covnames <- paste0("X", 1:p)
lamnames <- paste0("lam_", lam_grid)

slopes_q1 <- array(NA_real_, dim = c(n_rep, p, n_lam),
                  dimnames = list(NULL, covnames, lamnames))
conv_q1   <- matrix(TRUE, n_rep, n_lam)

for (r in seq_len(n_rep)) {
```

```

X1    <- rnorm(n)
Z     <- rnorm(n)
X2    <- 0.999 * X1 + sqrt(1 - 0.999^2) * Z
Xrest <- matrix(rnorm(n * (p - 2)), n, p - 2)
X     <- cbind(X1, X2, Xrest)

eps <- rnorm(n)
Y     <- 0.15 * X[, 1] + 0.15 * X[, 2] + 0.30 * X[, 3] + 0.15 * X[, 4] + eps

# Center covariates and standardize with divisor n (not n-1); center response.
Xc <- scale(X, center = TRUE, scale = FALSE)
sdn <- sqrt(colSums(Xc^2) / n)
Xs <- sweep(Xc, 2, sdn, "/")
Yc <- Y - mean(Y)

for (l in seq_len(n_lam)) {

  fit <- glmnet(Xs, Yc, alpha = 1, lambda = lam_grid[l],
               standardize = FALSE, intercept = FALSE,
               thresh = 1e-18, maxit = 1000000)
  b_std <- as.numeric(coef(fit))[-1]
  slopes_q1[r, , l] <- b_std / sdn
  conv_q1[r, l] <- (fit$jerr == 0)
}
}

cat("All", n_rep * n_lam, "lasso fits converged:", all(conv_q1), "\n")

```

All 800 lasso fits converged: TRUE

Part a

```

sel_q1      <- abs(slopes_q1) > 1e-8
sel_freq_q1 <- apply(sel_q1, c(2, 3), mean)
mabs_q1     <- apply(abs(slopes_q1), c(2, 3), mean)

kable(sel_freq_q1, digits = 3, caption = "Q1: Selection frequency (Lasso)")

```

Table 1: Q1: Selection frequency (Lasso)

	lam_0.08	lam_0.12	lam_0.18	lam_0.24
X1	0.480	0.445	0.390	0.335
X2	0.575	0.530	0.500	0.390
X3	0.985	0.965	0.910	0.700
X4	0.735	0.595	0.320	0.150
X5	0.355	0.165	0.065	0.015
X6	0.370	0.210	0.060	0.020
X7	0.350	0.205	0.065	0.005
X8	0.370	0.185	0.040	0.010
X9	0.390	0.210	0.105	0.010
X10	0.325	0.190	0.075	0.010

```
kable(mabs_q1, digits = 3, caption = "Q1: Mean absolute fitted slope, including zeros (Lasso)")
```

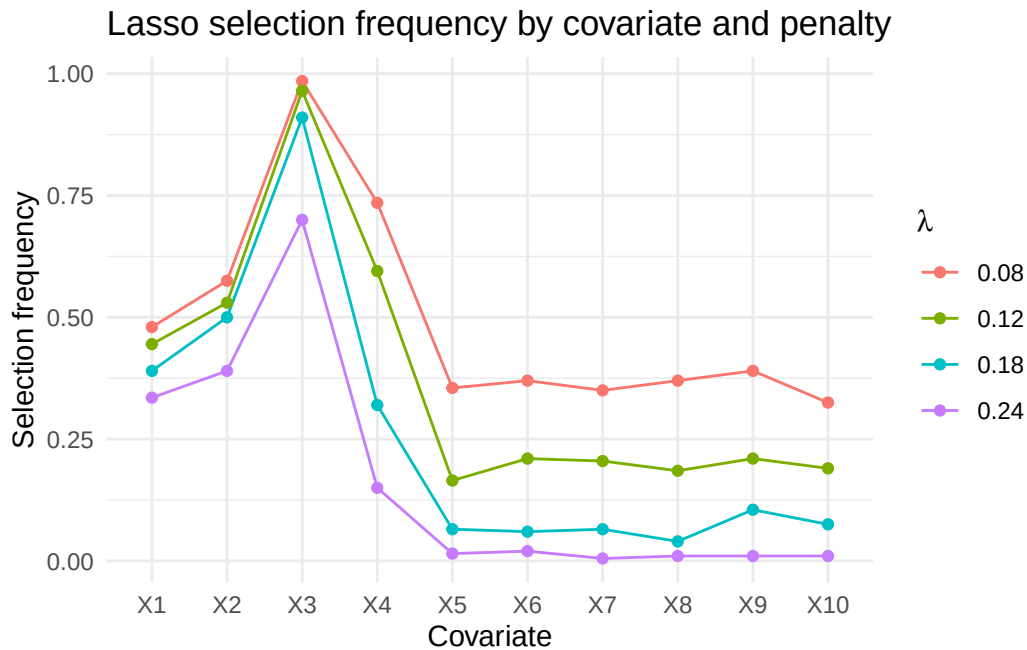
Table 2: Q1: Mean absolute fitted slope, including zeros (Lasso)

	lam_0.08	lam_0.12	lam_0.18	lam_0.24
X1	0.099	0.079	0.054	0.032
X2	0.112	0.092	0.063	0.037
X3	0.218	0.179	0.123	0.075
X4	0.075	0.048	0.021	0.007
X5	0.020	0.009	0.003	0.001
X6	0.022	0.010	0.003	0.001
X7	0.021	0.010	0.002	0.000
X8	0.019	0.008	0.001	0.000
X9	0.024	0.012	0.003	0.000
X10	0.018	0.009	0.002	0.000

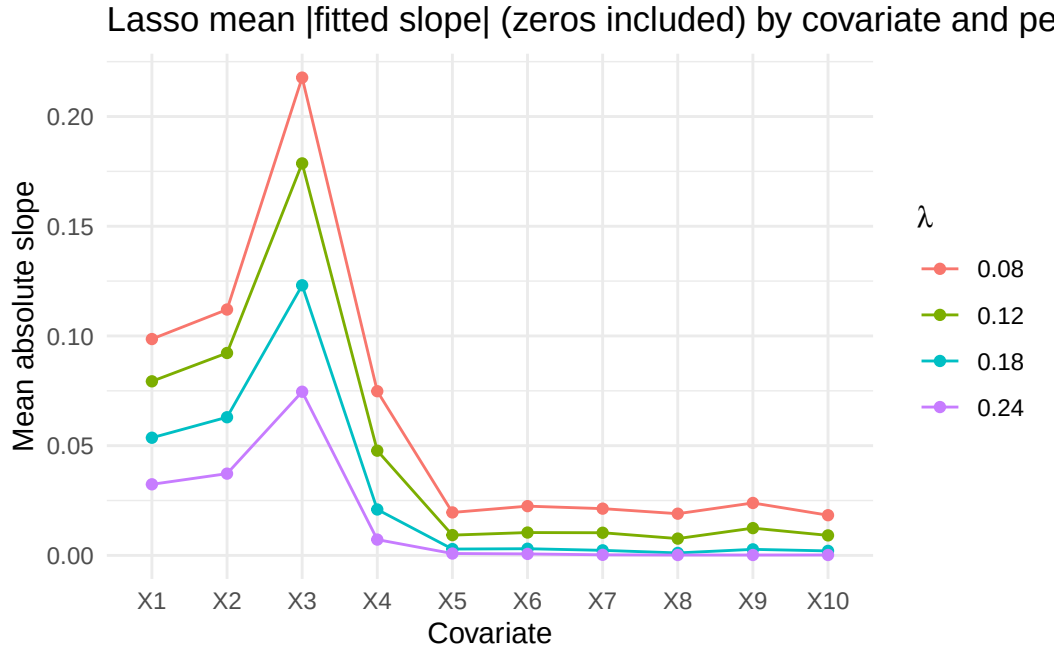
```
to_long <- function(mat, value_name) {
  df <- as.data.frame(mat)
  df$covariate <- factor(covnames, levels = covnames)
  pivot_longer(df, cols = all_of(lamnames), names_to = "lambda", values_to = value_name) |>
    mutate(lambda = factor(sub("lam_", "", lambda), levels = as.character(lam_grid)))
}

sel_long_q1 <- to_long(sel_freq_q1, "selection_freq")
mabs_long_q1 <- to_long(mabs_q1, "mean_abs_slope")
```

```
ggplot(sel_long_q1, aes(covariate, selection_freq, color = lambda, group = lambda)) +
  geom_line() + geom_point() +
  labs(title = "Lasso selection frequency by covariate and penalty",
        x = "Covariate", y = "Selection frequency", color = expression(lambda)) +
  theme_minimal()
```



```
ggplot(mabs_long_q1, aes(covariate, mean_abs_slope, color = lambda, group = lambda)) +
  geom_line() + geom_point() +
  labs(title = "Lasso mean |fitted slope| (zeros included) by covariate and penalty",
        x = "Covariate", y = "Mean absolute slope", color = expression(lambda)) +
  theme_minimal()
```



Part b

```
pair_outcomes_q1 <- t(sapply(seq_len(n_lam), function(l) {
  s1 <- sel_q1[, 1, l]; s2 <- sel_q1[, 2, l]
  c(only_X1 = mean(s1 & !s2),
    only_X2 = mean(!s1 & s2),
    both    = mean(s1 & s2),
    neither = mean(!s1 & !s2))
}))
rownames(pair_outcomes_q1) <- paste0("lambda=", lam_grid)
kable(pair_outcomes_q1, digits = 3, caption = "Q1b: (X1, X2) joint selection outcomes")
```

Table 3: Q1b: (X1, X2) joint selection outcomes

	only_X1	only_X2	both	neither
lambda=0.08	0.410	0.505	0.07	0.015
lambda=0.12	0.415	0.500	0.03	0.055
lambda=0.18	0.370	0.480	0.02	0.130
lambda=0.24	0.315	0.370	0.02	0.295

```
kable(cbind(freq_X1 = sel_freq_q1[1, ], freq_X2 = sel_freq_q1[2, ]),
       digits = 3, caption = "Q1b: marginal selection frequencies of X1 and X2")
```

Table 4: Q1b: marginal selection frequencies of X1 and X2

	freq_X1	freq_X2
lam_0.08	0.480	0.575
lam_0.12	0.445	0.530
lam_0.18	0.390	0.500
lam_0.24	0.335	0.390

Part c

```
noise_avg_q1 <- colMeans(sel_freq_q1[5:10, ])
kable(rbind(X3 = sel_freq_q1[3, ], X4 = sel_freq_q1[4, ], noise_avg = noise_avg_q1),
       digits = 3, caption = "Q1c: X3 and X4 selection frequency vs. average noise-covariate frequency")
```

Table 5: Q1c: X3 and X4 selection frequency vs. average noise-covariate frequency

	lam_0.08	lam_0.12	lam_0.18	lam_0.24
X3	0.985	0.965	0.910	0.700
X4	0.735	0.595	0.320	0.150
noise_avg	0.360	0.194	0.068	0.012

Question 2: Elastic net with an equal penalty mix

```
set.seed(43246)

slopes_q2 <- array(NA_real_, dim = c(n_rep, p, n_lam),
                  dimnames = list(NULL, covnames, lamnames))
conv_q2 <- matrix(TRUE, n_rep, n_lam)

for (r in seq_len(n_rep)) {
  X1 <- rnorm(n)
  Z <- rnorm(n)
```

```

X2    <- 0.999 * X1 + sqrt(1 - 0.999^2) * Z
Xrest <- matrix(rnorm(n * (p - 2)), n, p - 2)
X     <- cbind(X1, X2, Xrest)

eps <- rnorm(n)
Y    <- 0.15 * X[, 1] + 0.15 * X[, 2] + 0.30 * X[, 3] + 0.15 * X[, 4] + eps

Xc <- scale(X, center = TRUE, scale = FALSE)
sdn <- sqrt(colSums(Xc^2) / n)
Xs <- sweep(Xc, 2, sdn, "/")
Yc <- Y - mean(Y)

for (l in seq_len(n_lam)) {

  fit <- glmnet(Xs, Yc, family = gaussian(), alpha = 0.5, lambda = lam_grid[l],
               standardize = FALSE, intercept = FALSE,
               thresh = 1e-18, maxit = 1000000)
  b_std <- as.numeric(coef(fit))[-1]
  slopes_q2[r, , l] <- b_std / sdn
  conv_q2[r, l] <- (fit$jerr == 0)
}
}

cat("All", n_rep * n_lam, "elastic-net fits converged:", all(conv_q2), "\n")

```

All 800 elastic-net fits converged: TRUE

Part a

```

sel_q2    <- abs(slopes_q2) > 1e-8
sel_freq_q2 <- apply(sel_q2, c(2, 3), mean)

kable(sel_freq_q2, digits = 3, caption = "Q2: Selection frequency (Elastic net, alpha = 0.5)")

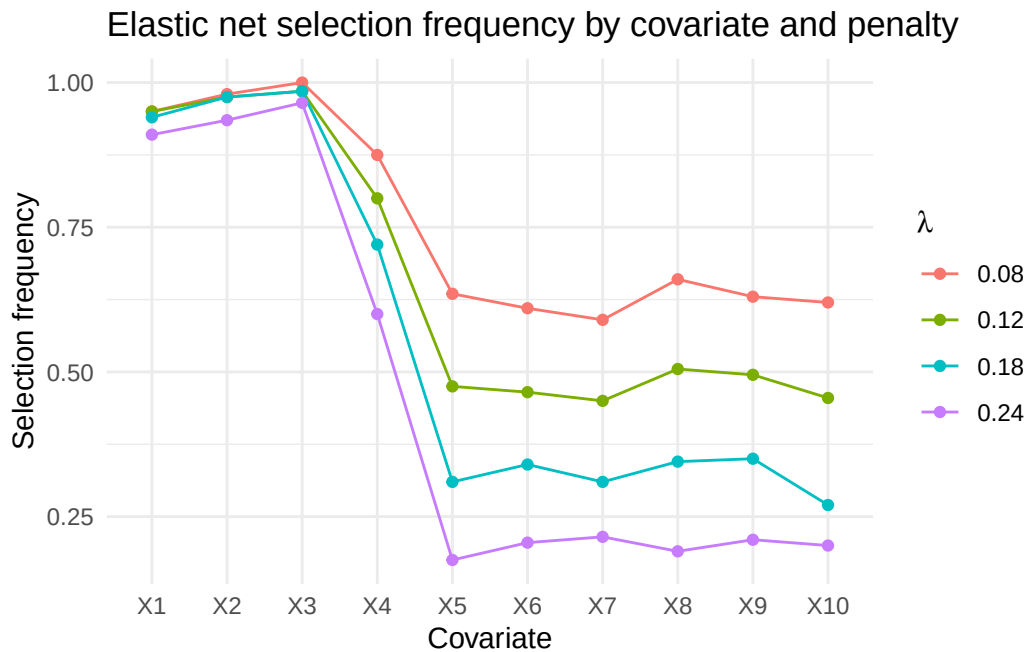
```

Table 6: Q2: Selection frequency (Elastic net, alpha = 0.5)

	lam_0.08	lam_0.12	lam_0.18	lam_0.24
X1	0.950	0.950	0.940	0.910
X2	0.980	0.975	0.975	0.935

	lam_0.08	lam_0.12	lam_0.18	lam_0.24
X3	1.000	0.985	0.985	0.965
X4	0.875	0.800	0.720	0.600
X5	0.635	0.475	0.310	0.175
X6	0.610	0.465	0.340	0.205
X7	0.590	0.450	0.310	0.215
X8	0.660	0.505	0.345	0.190
X9	0.630	0.495	0.350	0.210
X10	0.620	0.455	0.270	0.200

```
sel_long_q2 <- to_long(sel_freq_q2, "selection_freq")
ggplot(sel_long_q2, aes(covariate, selection_freq, color = lambda, group = lambda)) +
  geom_line() + geom_point() +
  labs(title = "Elastic net selection frequency by covariate and penalty",
       x = "Covariate", y = "Selection frequency", color = expression(lambda)) +
  theme_minimal()
```



```
kable(sel_freq_q2 - sel_freq_q1, digits = 3,
      caption = "Q2a: Elastic net minus Lasso selection frequency")
```

Table 7: Q2a: Elastic net minus Lasso selection frequency

	lam_0.08	lam_0.12	lam_0.18	lam_0.24
X1	0.470	0.505	0.550	0.575
X2	0.405	0.445	0.475	0.545
X3	0.015	0.020	0.075	0.265
X4	0.140	0.205	0.400	0.450
X5	0.280	0.310	0.245	0.160
X6	0.240	0.255	0.280	0.185
X7	0.240	0.245	0.245	0.210
X8	0.290	0.320	0.305	0.180
X9	0.240	0.285	0.245	0.200
X10	0.295	0.265	0.195	0.190

Part b

```

pair_outcomes_q2 <- t(sapply(seq_len(n_lam), function(l) {
  s1 <- sel_q2[, 1, 1]; s2 <- sel_q2[, 2, 1]
  c(only_X1 = mean(s1 & !s2),
    only_X2 = mean(!s1 & s2),
    both    = mean(s1 & s2),
    neither = mean(!s1 & !s2))
}))
rownames(pair_outcomes_q2) <- paste0("lambda=", lam_grid)
kable(pair_outcomes_q2, digits = 3, caption = "Q2b: (X1, X2) joint selection outcomes, elast.

```

Table 8: Q2b: (X1, X2) joint selection outcomes, elastic net

	only_X1	only_X2	both	neither
lambda=0.08	0.020	0.05	0.930	0.00
lambda=0.12	0.015	0.04	0.935	0.01
lambda=0.18	0.005	0.04	0.935	0.02
lambda=0.24	0.005	0.03	0.905	0.06

```

kable(pair_outcomes_q2 - pair_outcomes_q1, digits = 3,
  caption = "Q2b: Elastic net minus Lasso, joint outcome frequencies")

```

Table 9: Q2b: Elastic net minus Lasso, joint outcome frequencies

	only_X1	only_X2	both	neither
lambda=0.08	-0.390	-0.455	0.860	-0.015
lambda=0.12	-0.400	-0.460	0.905	-0.045
lambda=0.18	-0.365	-0.440	0.915	-0.110
lambda=0.24	-0.310	-0.340	0.885	-0.235

Part c

```
noise_avg_q2 <- colMeans(sel_freq_q2[5:10, ])
kable(rbind(elastic_net = noise_avg_q2, lasso = noise_avg_q1), digits = 3,
        caption = "Q2c: Average noise-covariate (X5..X10) selection frequency, by lambda")
```

Table 10: Q2c: Average noise-covariate (X5..X10) selection frequency, by lambda

	lam_0.08	lam_0.12	lam_0.18	lam_0.24
elastic_net	0.624	0.474	0.321	0.199
lasso	0.360	0.194	0.068	0.012

Question 3: Coordinate descent and the lasso path

```
set.seed(43247)

n3 <- 100
p3 <- 3
X3mat <- matrix(rnorm(n3 * p3), n3, p3)
eps3 <- rnorm(n3)
Y3 <- 1.5 + X3mat[, 1] - 2 * X3mat[, 2] + eps3

Yc3 <- Y3 - mean(Y3)
Xc3 <- scale(X3mat, center = TRUE, scale = FALSE)
sdn3 <- sqrt(colSums(Xc3^2) / n3)
Xs3 <- sweep(Xc3, 2, sdn3, "/")

soft_threshold <- function(a, lambda) sign(a) * pmax(abs(a) - lambda, 0)
```

```

coord_descent <- function(X, y, lambda, beta_init, tol = 1e-8, max_iter = 10000) {
  p <- ncol(X)
  beta <- beta_init
  converged <- FALSE
  for (iter in seq_len(max_iter)) {
    beta_old <- beta
    for (j in seq_len(p)) {
      r_j <- y - X[, -j, drop = FALSE] %*% beta[-j]
      a_j <- mean(X[, j] * r_j)
      beta[j] <- soft_threshold(a_j, lambda)
    }
    if (max(abs(beta - beta_old)) < tol) { converged <- TRUE; break }
  }
  list(beta = as.numeric(beta), converged = converged, iters = iter)
}

lambda_path <- c(2.5, 2.0, 1.5, 1.0, 0.6, 0.4, 0.25, 0.15, 0.08, 0.03)
beta_path <- matrix(NA_real_, length(lambda_path), p3)
conv_path <- logical(length(lambda_path))
beta_cur <- rep(0, p3)
for (k in seq_along(lambda_path)) {
  res <- coord_descent(Xs3, Yc3, lambda_path[k], beta_cur)
  beta_path[k, ] <- res$beta
  conv_path[k] <- res$converged
  beta_cur <- res$beta
}
colnames(beta_path) <- c("beta1", "beta2", "beta3")
rownames(beta_path) <- paste0("lambda=", lambda_path)

```

Part a

```

kable(beta_path, digits = 4, caption = "Q3a: Coordinate-descent lasso path")

```

Table 11: Q3a: Coordinate-descent lasso path

	beta1	beta2	beta3
lambda=2.5	0.0000	0.0000	0
lambda=2	0.0000	0.0000	0
lambda=1.5	0.0000	-0.3272	0

	beta1	beta2	beta3
lambda=1	0.0000	-0.8272	0
lambda=0.6	0.2835	-1.2495	0
lambda=0.4	0.5006	-1.4666	0
lambda=0.25	0.6634	-1.6294	0
lambda=0.15	0.7719	-1.7380	0
lambda=0.08	0.8479	-1.8140	0
lambda=0.03	0.9022	-1.8682	0

```
cat("All fits converged:", all(conv_path), "\n")
```

All fits converged: TRUE

```
lambda_max <- max(abs(colMeans(Xs3 * Yc3)))
cat("lambda_max =", round(lambda_max, 4), "\n")
```

lambda_max = 1.8272

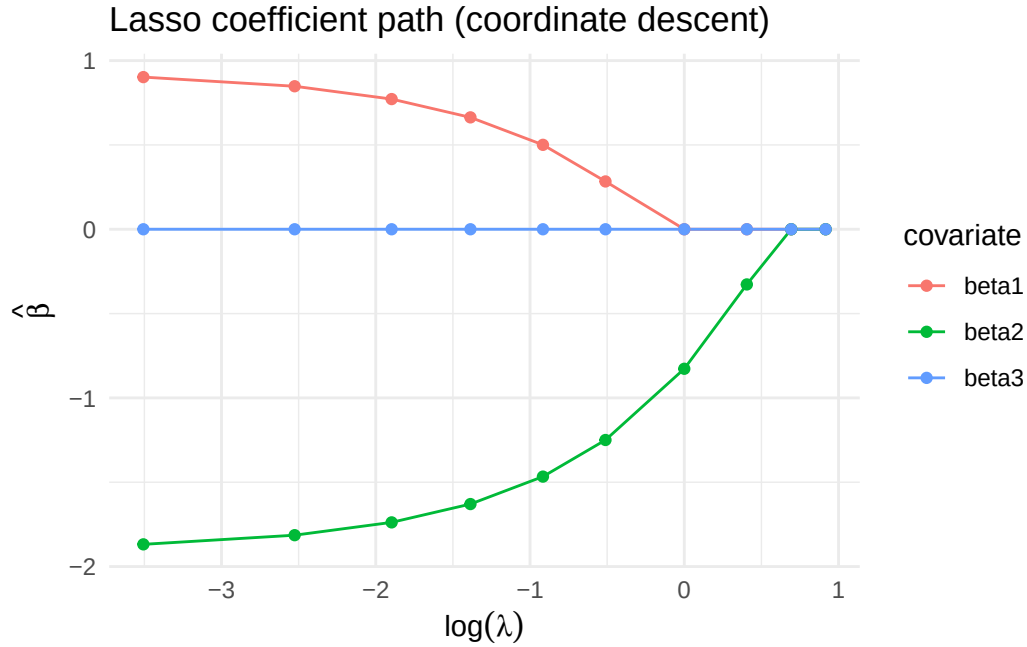
```
cat("All slopes zero for lambda >= lambda_max:",
    all(beta_path[lambda_path >= lambda_max, ] == 0), "\n")
```

All slopes zero for lambda >= lambda_max: TRUE

Part b

```
path_df <- data.frame(log_lambda = log(lambda_path), beta_path)
path_long <- pivot_longer(path_df, cols = starts_with("beta"),
                          names_to = "covariate", values_to = "beta_hat")

ggplot(path_long, aes(log_lambda, beta_hat, color = covariate)) +
  geom_line() + geom_point() +
  labs(title = "Lasso coefficient path (coordinate descent)",
       x = expression(log(lambda)), y = expression(hat(beta))) +
  theme_minimal()
```



Interpretation. X_2 , with the largest true coefficient in magnitude (-2), enters the path first (at the largest λ below $\lambda_{\max}$), followed by X_1 (true coefficient 1); larger true signal produces a larger score a_j at any given point on the path, so it crosses the soft-threshold cutoff sooner as λ decreases from $\lambda_{\max}$.

Part c

```
ols_fit <- lm(Yc3 ~ Xs3 - 1)
kable(rbind(lasso_smallest_lambda = beta_path[nrow(beta_path), ],
            ols = coef(ols_fit)),
      digits = 4, caption = "Q3c: Smallest-lambda lasso fit vs. OLS on the same standardized data")
```

Table 12: Q3c: Smallest-lambda lasso fit vs. OLS on the same standardized data

	beta1	beta2	beta3
lasso_smallest_lambda	0.9022	-1.8682	0.0000
ols	0.9329	-1.9016	0.0088

Question 4: Tuning and comparing penalized regressions for diabetes prediction

```
diabetes <- read.csv("~/Desktop/stat432-fall2026/week-04/data/diabetes.csv")
split    <- read.csv("~/Desktop/stat432-fall2026/week-04/data/diabetes-split-folds.csv")

stopifnot(nrow(diabetes) == nrow(split), all(split$row_id == seq_len(nrow(split))))

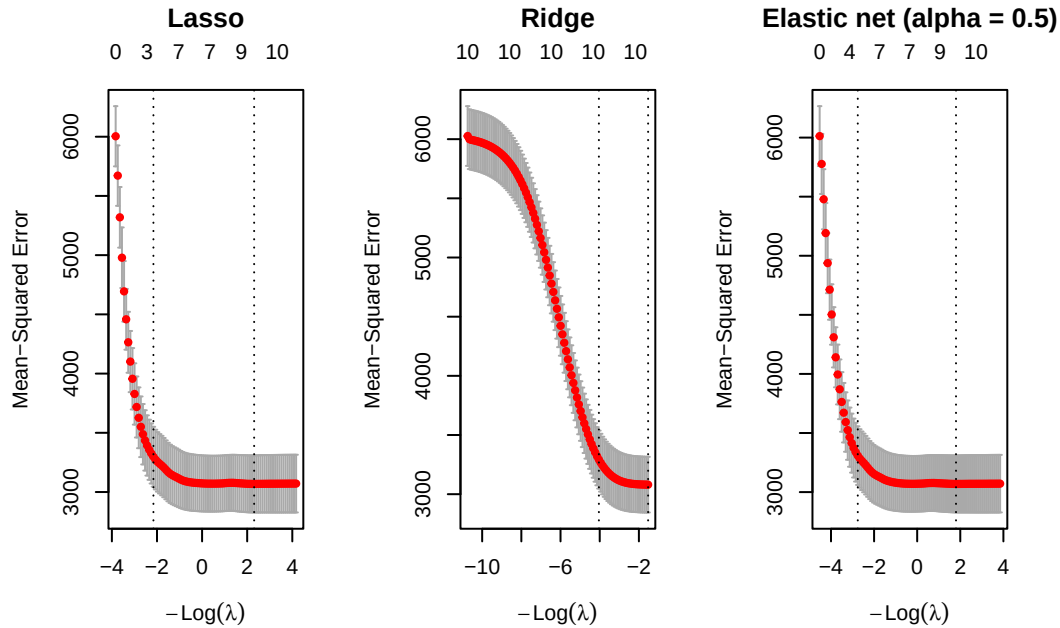
X_all <- as.matrix(diabetes[, setdiff(names(diabetes), "y")])
y_all <- diabetes$y

train_rows <- split$split == "train"
test_rows  <- split$split == "test"
foldid     <- split$cv_fold[train_rows]

X_train <- X_all[train_rows, ]; y_train <- y_all[train_rows]
X_test  <- X_all[test_rows, ]; y_test  <- y_all[test_rows]

cv_lasso <- cv.glmnet(X_train, y_train, alpha = 1, foldid = foldid)
cv_ridge <- cv.glmnet(X_train, y_train, alpha = 0, foldid = foldid)
cv_enet  <- cv.glmnet(X_train, y_train, alpha = 0.5, foldid = foldid)

par(mfrow = c(1, 3))
plot(cv_lasso); title("Lasso", line = 2.5)
plot(cv_ridge); title("Ridge", line = 2.5)
plot(cv_enet); title("Elastic net (alpha = 0.5)", line = 2.5)
```



```
par(mfrow = c(1, 1))

cv_table <- data.frame(
  method      = c("lasso", "ridge", "elastic net"),
  lambda_min  = c(cv_lasso$lambda.min, cv_ridge$lambda.min, cv_enet$lambda.min),
  cv_error    = c(min(cv_lasso$cvm), min(cv_ridge$cvm), min(cv_enet$cvm))
)
kable(cv_table, digits = 4, caption = "Q4: Selected penalty and CV error by method")
```

Table 13: Q4: Selected penalty and CV error by method

method	lambda_min	cv_error
lasso	0.0995	3068.424
ridge	4.6163	3080.365
elastic net	0.1651	3069.013

```
best      <- cv_table[which.min(cv_table$cv_error), ]
alpha_map <- c(lasso = 1, ridge = 0, "elastic net" = 0.5)
final_fit <- glmnet(X_train, y_train, alpha = alpha_map[[best$method]], lambda = best$lambda)
pred_test <- predict(final_fit, newx = X_test)
test_mse  <- mean((y_test - pred_test)^2)
```

```
cat("Selected method:", best$method, "\n")
```

Selected method: lasso

```
cat("Selected lambda:", best$lambda_min, "\n")
```

Selected lambda: 0.09945587

```
cat("Test MSE:", test_mse, "\n")
```

Test MSE: 2731.794